



scRNA-seq Data Analysis



ELIXIR Single-Cell
Omics Community



Swiss Institute of
Bioinformatics



Funded by ELIXIR Europe staff exchange programme

The Staff Exchange Programme supports defined, goal-oriented activities that contribute to capacity building within ELIXIR Nodes and strengthen links across the ELIXIR community.

<https://elixir-europe.org/internal-projects/staff-exchange-programme>



Ongoing collaboration



2025

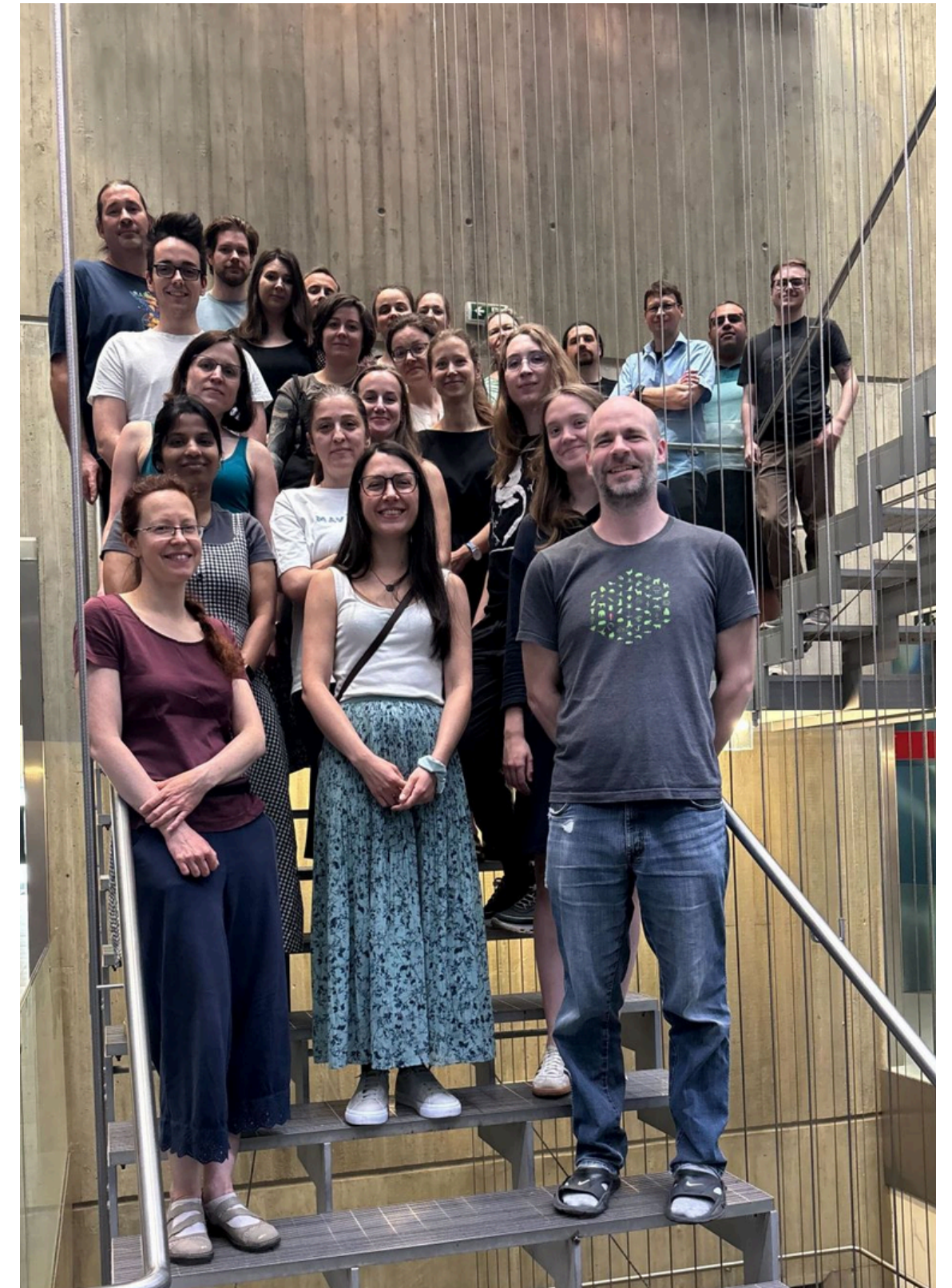
Statistics for Life Sciences in R Course



<https://www.elixir-czech.cz/events/python-basics-05-2026>
<https://www.elixir-czech.cz/events/statistics-for-life-sciences-in-r-course>
<https://elixir-europe.org/internal-projects/staff-exchange-programme/bioinformatics-stats>

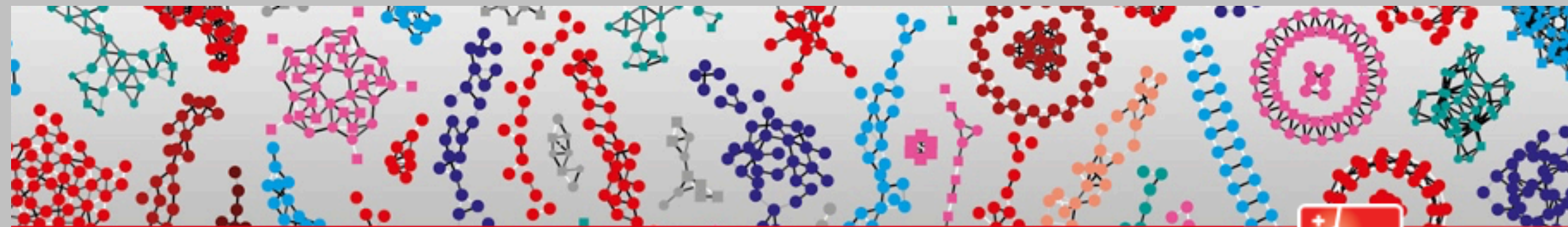
2026

Python Basics



Course on scRNA-seq Data Analysis

Based on the course by SIB <https://sib-swiss.github.io/single-cell-r-training/>



Single cell transcriptomics

Rachel Marcone Jeitziner

Tania Wyss

Alex Lederer

Geert van Geest



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Single-Cell Omics
Community

Course on scRNA-seq Data Analysis

- Day 1 – Introduction and QC
- Day 2 – Data processing and visualisation
- Day 3 – Annotation and interpretation
- Day 4 - Final touches of group work and presentation
- Day 1-4 – Group work

https://elixir.ut.ee/news/2026/06/07/scRNA-seq_Data_Analysis/



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Course on scRNA-seq Data Analysis

Learning outcomes

- Explain the **characteristics** of the most-used **methods** to scRNA-seq data
- Perform **quality control** at different steps of the analysis
- Use **dimensionality reduction** (PCA, t-SNE, UMAP)
- Apply **best practices** for **translating** single cell transcriptomic data into **biological knowledge**



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Course on scRNA-seq Data Analysis

Trainers CZ

Lucie Pfeiferová (IMG)
Jan Kubovčíak (IMG)
Vojtěch Melichar (IMG)



Organisers EE

Diana Pilvar (UT)
Priit Adler (UT)



Course on scRNA-seq Data Analysis

Communication

- in Person
- GitHub page (bioinfocz.github.io/Course-on-scRNA-seq-data-analysis)
- Google docs



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Course on scRNA-seq Data Analysis

Day 1 – Monday – Introduction and QC

9:00 – 9:30 Introduction

9:30 – 10:30 Introduction to scRNA-seq

10:30 – 11:00 Break

11:00 – 12:30 10× and Cellranger

12:30 – 13:30 Lunch

13:30 – 15:00 Analysis tools and QC

15:00 – 15:30 Break

15:30 – 17:00 Group work



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Projects (group work)

- 4 projects
- read the project page on GitHub pages
- assign yourself to the project you would like
- till today after lunch
- first come first serve basis
- QR code to assign:



Course on scRNA-seq Data Analysis

- Cable connection is preferred
 - Eduroam WiFi is preferred
 - Free WiFi is an alternative
- AC sockets are below the desks
- Please sign the attendance list
- Any technical difficulties?
- Any specific requirements?

Introduction round

- Your name
- What do you do?
- Why you join the course